Customer Segmentation & Revenue Analysis Project

Tools Used:

Excel • Power BI • DAX

Project Type:

Customer Segmentation/RFM Segmentation

Prepared By:

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2026

Project Overview

This report summarizes insights derived from the Customer Segmentation Project. It focuses on identifying which customer segments contribute the most revenue and what strategies should be promoted to maximize the profitability. Excel, Power BI & DAX functions are used to analyze and finalize the project.

Business Problem

Businesses often face problems in identifying which customer segment is bringing in revenue and which segment can pose a revenue risk due to inactivity. Identifying high-value customers and monitoring their engagement level is essential and crucial for maintaining a stable growth.

The Objective of this project is to analyze the customer purchasing behavior and bucketing them according to their spendings, activity and Average Order Value (AOV).

Dataset Description

The dataset contains customer level information including Spending Behavior, Income band, spend score and engagement metrics which will help us analyze different customer segments. Each row represents an individual customer with attributes describing their spending behavior and activity levels.

Key attributes included in the dataset:

- Customer ID
- Income Band
- Spend Behavior (Low / Medium / High / Very High)
- Recency Band
- Total Orders
- Average Order Value (AOV)
- Last purchase days ago
- Revenue per Customer

The dataset enables segmentation analyses and helps us bucket customers based on their income levels and spending behavior and how these impact overall revenue contribution.

Methodology

This analysis was done with the combination of Excel for Data Preparation and Power BI for Data Modeling and Data Visualization.

The analytical approach included:

1. Data Preparation:
   Initial data cleaning and structuring were performed in Excel to ensure the data is clean and ready for analysis.

2. Customer Segmentaion:
   Customers were segmented based on:
   - Income Band
   - Spending Behavior
   - Recency Band / Purchase Activity

3. RFM based Analysis:
   Customer behavior was analyzed using key RFM strategies:
   - Recency – How recently customers made a purchase.
   - Frequency – How frequently customers placed orders.
   - Monetary – How much revenue each customer generated.

4. Dashboard Creation:
   Interactive dashboards were created in Power BI to visualize:
   - Revenue contribution per customer
   - Total customers in each income, spending and recent activity segments
   - Customer inactivity risk
   - Revenue contribution by Income band and Spending Behavior.

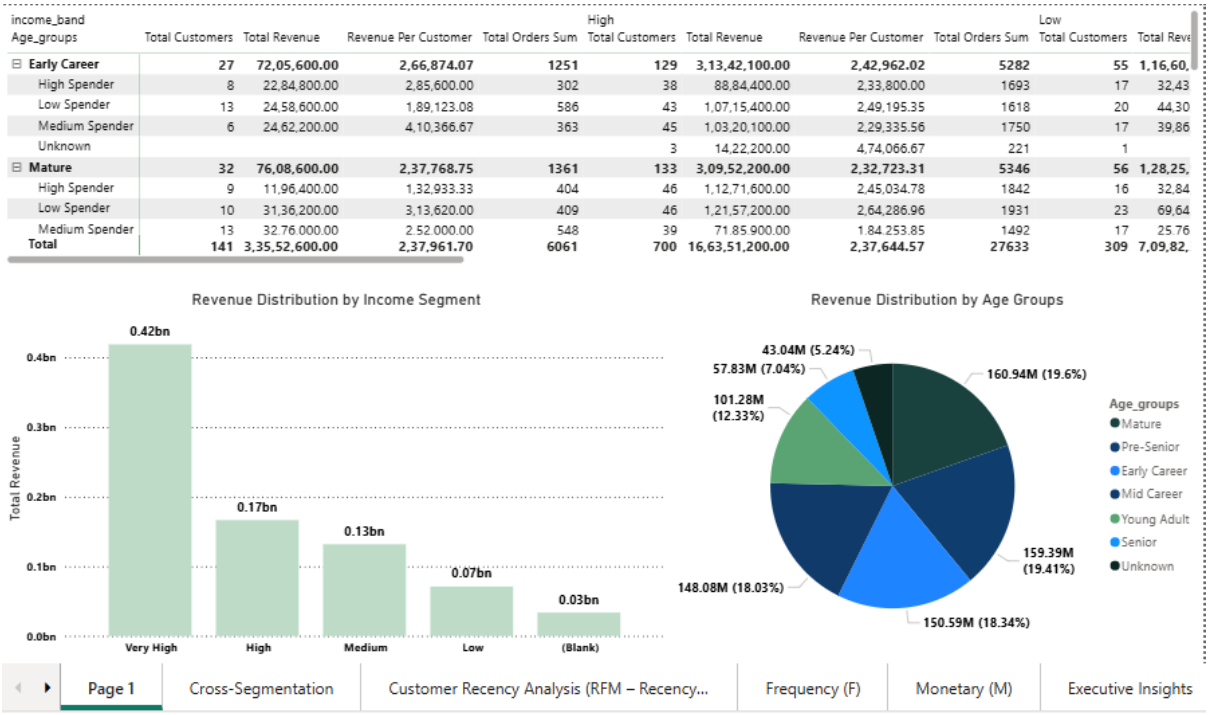
Dashboard Overview

An interactive Power BI dashboard was developed showing different visualizations to analyze customer segments and identify key findings and revenue drivers. The dashboard allows users to explore different revenue distributions across different Income Bands, Spending behaviors and customer activity levels.

The dashboard is divided into multiple sections:

- Revenue Overview
- Customer Segmentation Analysis
- Recency Analysis
- Frequency Analysis
- Monetary value Analysis
- Executive Summary

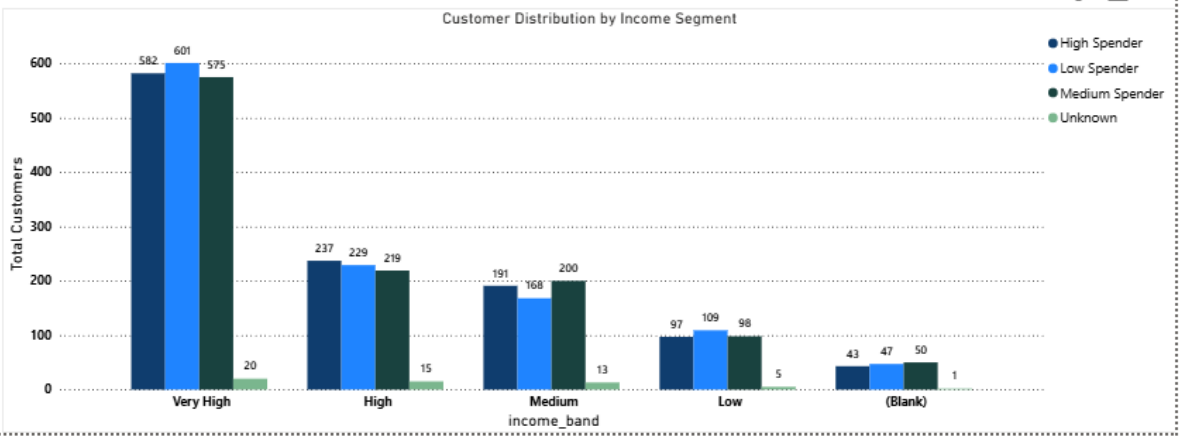
These Dashboards helps highlight customer groups that generate the most revenue while also identifying which customer segments poses possible retention risk.



income_band / Age_groups	Total Customers	Total Revenue	Revenue Per Customer	Total Orders Sum	High Total Customers	High Total Revenue	High Revenue Per Customer	High Total Orders Sum	Low Total Customers	Low Total Revenue
Early Career	**27**	**72,05,600.00**	**2,66,874.07**	**1251**	**129**	**3,13,42,100.00**	**2,42,962.02**	**5282**	**55**	**1,16,60,**
High Spender	8	22,84,800.00	2,85,600.00	302	38	88,84,400.00	2,33,800.00	1693	17	32,43
Low Spender	13	24,58,600.00	1,89,123.08	586	43	1,07,15,400.00	2,49,195.35	1618	20	44,30
Medium Spender	6	24,62,200.00	4,10,366.67	363	45	1,03,20,100.00	2,29,335.56	1750	17	39,86
Unknown					3	14,22,200.00	4,74,066.67	221	1	
Mature	**32**	**76,08,600.00**	**2,37,768.75**	**1361**	**133**	**3,09,52,200.00**	**2,32,723.31**	**5346**	**56**	**1,28,25,**
High Spender	9	11,96,400.00	1,32,933.33	404	46	1,12,71,600.00	2,45,034.78	1842	16	32,84
Low Spender	10	31,36,200.00	3,13,620.00	409	46	1,21,57,200.00	2,64,286.96	1931	23	69,64
Medium Spender	13	32,76,000.00	2,52,000.00	548	39	71,85,900.00	1,84,253.85	1492	17	25,76
Total	**141**	**3,35,52,600.00**	**2,37,961.70**	**6061**	**700**	**16,63,51,200.00**	**2,37,644.57**	**27633**	**309**	**7,09,82,**

Page 1 shows that Revenue is highly concentrated within the Very High income segment with 50.98 %. Additionally, The Mature Age group contributes 19.6% of overall revenue.

spend_behavior	High Spender			Low Spender			Medium Spender			U
income_band	Total Customers	Revenue Per Customer	Sum of spend_score	Total Customers	Revenue Per Customer	Sum of spend_score	Total Customers	Revenue Per Customer	Sum of spend_score	Tc
	43	2,04,713.95	3571	47	2,77,521.28	785	50	2,21,206.00	2432	
High	237	2,49,026.16	19953	229	2,29,436.24	4039	219	2,28,936.99	11025	
Low	97	2,06,291.75	8105	109	2,59,131.19	1922	98	2,30,475.51	4998	
Medium	191	2,23,679.58	15962	168	2,33,497.02	2860	200	2,33,215.50	10092	
Very High	582	2,40,085.57	48358	601	2,32,060.23	10558	575	2,35,926.43	28421	
Total	**1150**	**2,35,030.26**	**95949**	**1154**	**2,36,157.19**	**20164**	**1142**	**2,32,999.04**	**56968**	

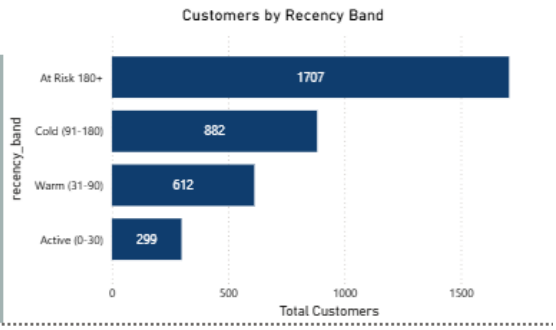


Cross-Segmentation | Customer Recency Analysis (RFM – Recency... | Frequency (F) | Monetary (M) | Executive Insights | Strat

Cross-Segmentation visualizes the distribution of customers Spend Behaviour across different Income Segment. It concludes that **Low Spender** in the **Very High** income Band form the largest customer group, with 601 customers, indicating a significant opportunity to increase revenue through targeted upselling strategies.

recency_band	Active (0-30)		At Risk 180+		Cold (91-180)		Warm (31-90)		Total	
income_band	Total Customers	Revenue Per Customer	Total Customers	Revenue Per Customer	Total Customers	Revenue Per Customer	Total Customers	Revenue Per Customer	Total Customers	F
	10	**2,85,170.00**	**67**	**2,18,029.85**	**40**	**2,45,260.00**	**24**	**2,61,770.83**	**141**	
High Spender	5	3,10,960.00	22	1,93,486.36	12	2,04,300.00	4	1,34,900.00	**43**	
Low Spender	4	3,18,525.00	19	2,38,536.84	12	2,97,841.67	12	3,05,258.33	**47**	
Medium Spender	1	22,800.00	26	2,23,811.54	16	2,36,543.75	7	2,04,814.29	**50**	
Unknown							1	6,46,100.00	**1**	
High	**57**	**2,50,984.21**	**333**	**2,44,106.61**	**183**	**2,36,216.94**	**127**	**2,16,770.87**	**700**	
High Spender	23	2,61,643.48	107	2,77,152.34	65	2,20,898.46	42	2,13,992.86	**237**	
Low Spender	15	2,65,220.00	111	2,14,826.13	59	2,55,357.63	44	2,19,336.36	**229**	
Medium Spender	17	2,01,547.06	108	2,33,317.59	56	2,35,816.07	38	2,18,602.63	**219**	
Unknown	2	4,41,850.00	7	3,69,742.86	3	1,99,166.67	3	1,94,833.33	**15**	
Low	**28**	**2,31,882.14**	**148**	**2,19,033.11**	**73**	**2,33,843.84**	**60**	**2,50,035.00**	**309**	
High Spender	8	1,64,900.00	46	2,18,391.30	28	2,25,982.14	15	1,54,506.67	**97**	
Low Spender	11	2,49,009.09	50	2,54,190.00	24	2,62,720.83	24	2,70,475.00	**109**	
Total	**299**	**2,29,311.04**	**1707**	**2,36,753.95**	**882**	**2,32,524.04**	**612**	**2,34,241.01**	**3500**	

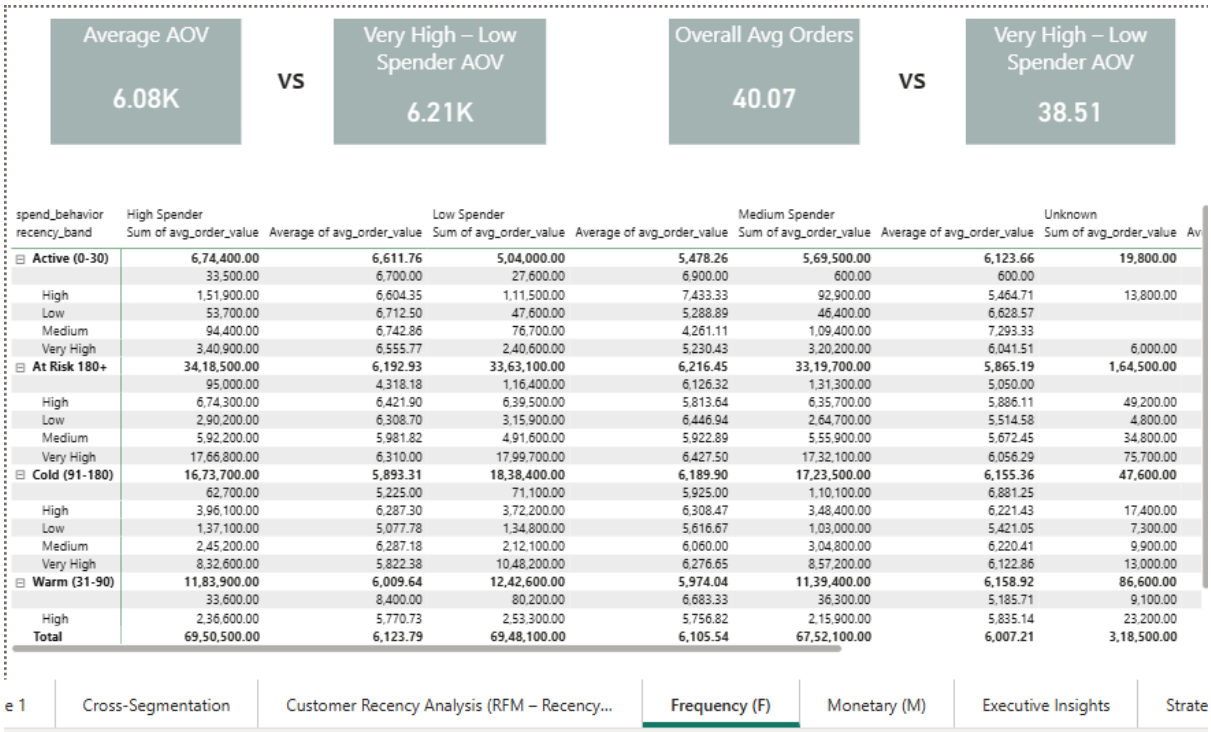
Very High income_band	Active (0-30) recency_band	46 Total Customers	2,16,917.39 Revenue Per Custo...
Very High income_band	At Risk 180+ recency_band	285 Total Customers	2,32,185.61 Revenue Per Custo...
Very High income_band	Cold (91-180) recency_band	168 Total Customers	2,42,641.67 Revenue Per Custo...
Very High income_band	Warm (31-90) recency_band	102 Total Customers	2,21,110.78 Revenue Per Custo...



Cross-Segmentation | Customer Recency Analysis (RFM – Recenc... | Frequency (F) | Monetary (M) | Executive Insights | Str

Customer Recency Analysis (R) analyses customer activity based on the number of days since their last purchase. The chart shows that the majority of customers fall into the At Risk (180+ days) category with 1707 customers, indicating that a large portion of the customer base has been inactive for a long time.

In comparison with Active (0-30 days) segment which has only 299 customers, suggesting in low engagement of customers.



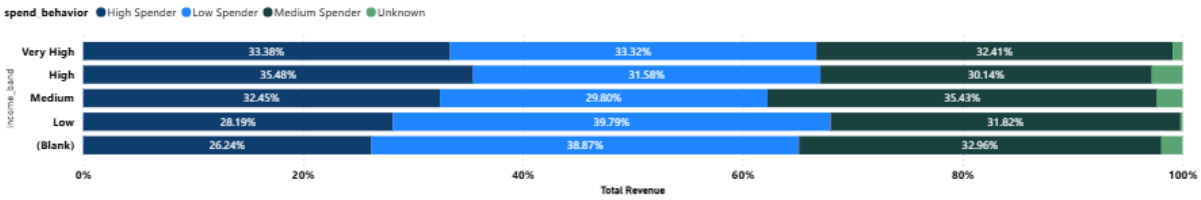
spend_behavior	High Spender		Low Spender		Medium Spender		Unknown	
recency_band	Sum of avg_order_value	Average of avg_order_value	Sum of avg_order_value	Average of avg_order_value	Sum of avg_order_value	Average of avg_order_value	Sum of avg_order_value	Av
⊟ **Active (0-30)**	**6,74,400.00**	**6,611.76**	**5,04,000.00**	**5,478.26**	**5,69,500.00**	**6,123.66**	**19,800.00**	
	33,500.00	6,700.00	27,600.00	6,900.00	600.00	600.00		
High	1,51,900.00	6,604.35	1,11,500.00	7,433.33	92,900.00	5,464.71	13,800.00	
Low	53,700.00	6,712.50	47,600.00	5,288.89	46,400.00	6,628.57		
Medium	94,400.00	6,742.86	76,700.00	4,261.11	1,09,400.00	7,293.33		
Very High	3,40,900.00	6,555.77	2,40,600.00	5,230.43	3,20,200.00	6,041.51	6,000.00	
⊟ **At Risk 180+**	**34,18,500.00**	**6,192.93**	**33,63,100.00**	**6,216.45**	**33,19,700.00**	**5,865.19**	**1,64,500.00**	
	95,000.00	4,318.18	1,16,400.00	6,126.32	1,31,300.00	5,050.00		
High	6,74,300.00	6,421.90	6,39,500.00	5,813.64	6,35,700.00	5,886.11	49,200.00	
Low	2,90,200.00	6,308.70	3,15,900.00	6,446.94	2,64,700.00	5,514.58	4,800.00	
Medium	5,92,200.00	5,981.82	4,91,600.00	5,922.89	5,55,900.00	5,672.45	34,800.00	
Very High	17,66,800.00	6,310.00	17,99,700.00	6,427.50	17,32,100.00	6,056.29	75,700.00	
⊟ **Cold (91-180)**	**16,73,700.00**	**5,893.31**	**18,38,400.00**	**6,189.90**	**17,23,500.00**	**6,155.36**	**47,600.00**	
	62,700.00	5,225.00	71,100.00	5,925.00	1,10,100.00	6,881.25		
High	3,96,100.00	6,287.30	3,72,200.00	6,308.47	3,48,400.00	6,221.43	17,400.00	
Low	1,37,100.00	5,077.78	1,34,800.00	5,616.67	1,03,000.00	5,421.05	7,300.00	
Medium	2,45,200.00	6,287.18	2,12,100.00	6,060.00	3,04,800.00	6,220.41	9,900.00	
Very High	8,32,600.00	5,822.38	10,48,200.00	6,276.65	8,57,200.00	6,122.86	13,000.00	
⊟ **Warm (31-90)**	**11,83,900.00**	**6,009.64**	**12,42,600.00**	**5,974.04**	**11,39,400.00**	**6,158.92**	**86,600.00**	
	33,600.00	8,400.00	80,200.00	6,683.33	36,300.00	5,185.71	9,100.00	
High	2,36,600.00	5,770.73	2,53,300.00	5,756.82	2,15,900.00	5,835.14	23,200.00	
Total	**69,50,500.00**	**6,123.79**	**69,48,100.00**	**6,105.54**	**67,52,100.00**	**6,007.21**	**3,18,500.00**	

Frequency Analysis (F) shows that the overall average order value (AOV) is about 6.08K, while the Very High Income – Low Spender segment has a slightly higher AOV of around 6.21K.
The average number of orders per customer is around 40, which is similar to the Very High Income – Low Spender segment, indicating comparable purchasing frequency across these groups.

spend_behavior	High Spender		Low Spender		Medium Spender		Unknown	
income_band	%GT Total Customers	%GT Total Revenue	%GT Total Customers	%GT Total Revenue	%GT Total Customers	%GT Total Revenue	%GT Total Customers	%GT Tot
	1.23%	1.07%	1.34%	1.59%	1.43%	1.35%	0.03%	
High	6.77%	7.19%	6.54%	6.40%	6.26%	6.11%	0.43%	
Low	2.77%	2.44%	3.11%	3.44%	2.80%	2.75%	0.14%	
Medium	5.46%	5.20%	4.80%	4.78%	5.71%	5.68%	0.37%	
Very High	16.63%	17.02%	17.17%	16.98%	16.43%	16.52%	0.57%	
Total	**32.86%**	**32.92%**	**32.97%**	**33.19%**	**32.63%**	**32.40%**	**1.54%**	

spend_behavior	High Spender		Low Spender		Medium Spender		Unknown		**Total**
income_band	Total Customers	Total Revenue	Total Customers	Total Revenue	Total Customers	Total Revenue	Total Customers	Total Revenue	**Total Customers**
	43	88,02,700.00	47	1,30,43,500.00	50	1,10,60,300.00	1	6,46,100.00	**141**
High	237	5,90,19,200.00	229	5,25,40,900.00	219	5,01,37,200.00	15	46,53,900.00	**700**
Low	97	2,00,10,300.00	109	2,82,45,300.00	98	2,25,86,600.00	5	1,40,100.00	**309**
Medium	191	4,27,22,800.00	168	3,92,27,500.00	200	4,66,43,100.00	13	30,49,400.00	**572**
Very High	582	13,97,29,800.00	601	13,94,68,200.00	575	13,56,57,700.00	20	37,60,100.00	**1778**
Total	**1150**	**27,02,84,800.00**	**1154**	**27,25,25,400.00**	**1142**	**26,60,84,900.00**	**54**	**1,22,49,600.00**	**3500**

Total Revenue by income_band and spend_behavior



age 1 | Cross-Segmentation | Customer Recency Analysis (RFM – Recency... | Frequency (F) | Monetary (M) | Executive Insights | Strat

Monetary analysis (M) shows that **Very High Income** customers generate the highest total revenue across segments.
Within this group, **Low Spenders** are the largest segment (601 customers), suggesting a potential opportunity for upselling.

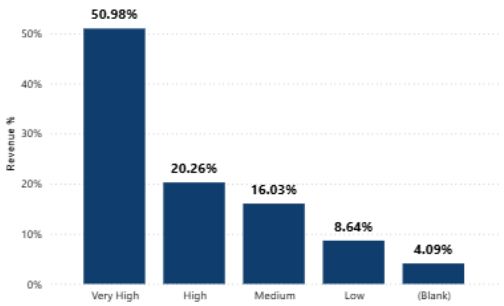
821.14M
Total Revenue

50.98%
z % Revenue Very High

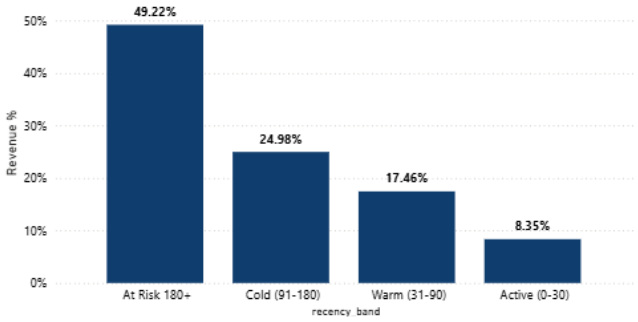
48.77%
z % Customers At Risk

49.22%
z % revenue from At Risk

Revenue % by income_band



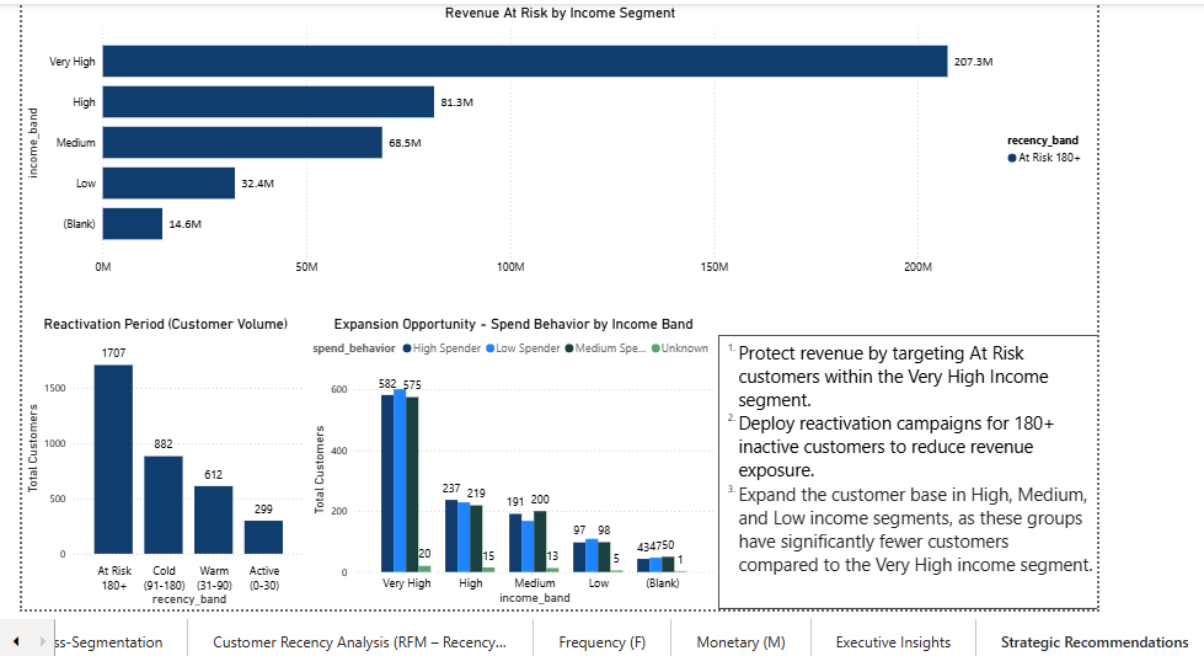
Revenue % by recency_band



The company generates 821.14M in revenue, with 50.98% concentrated in the Very High Income segment. Furthermore, 49.22% of total revenue is tied to customers inactive for more than 180 days, indicating significant retention exposure.

Page 1 | Cross-Segmentation | Customer Recency Analysis (RFM – Recency... | Frequency (F) | Monetary (M) | **Executive Insights** | Strateg

In **Executive Insights**, Total revenue is **821.14M**, with 50.98% generated by **Very High Income** customers, indicating strong revenue concentration. At the same time, **48.77%** of customers are **At Risk**, contributing **49.22%** of total revenue, which highlights a major customer retention challenge.



In **Strategic Recommendations,** we found that **Very High** income segment contributed the most revenue with **207.3M** in revenue. Additionally, customer count in **High**, **Medium** and **Low** spend behaviour is significantly low in comparison to the **Very High** income segment, suggesting the expansion in these spending segments is crucial and could help diversify revenue.

Key Insights

- Approximate revenue of the company generated is **821.14M**, with **50.98** of the revenue coming from the Very High income segment, showing strong revenue concentration in this segment.
- 48.77% customers of the total are in the At Risk (Inactive for 180+ days) Recency segment.

- Around 49.22% of the total revenue comes from the At Risk (Inactive for 180+ days) Recency segment.
- The **Very High** income segment has the largest customer count, having around 600 customer per each spending segment.
- In comparison, the highest customer count in the other three segments being Low, Medium & High is around 237, across all spending groups. This makes customer concentration significantly lower in these groups.
- This highlights that, the company relies heavily on the Very High income segment, creating an opportunity to expand customer base in other groups to generate additional revenue opportunities.

Business Recommendations

- **Segment-Based Marketing**
  Use customer segmentation insights to create more personalized marketing strategies best suited to different Income and Spending segments.
- **Customer Retention Strategy**
  Implement targeting campaigns, to call back the At Risk (180+ days) customers, as they represent nearly 50% of the total revenue.
- **Revenue Growth Opportunities**
  Encourage higher spending among **Low** and **Medium spenders**, particularly within higher income segments, through targeted promotions or personalized offers.
- **Customer Base Expansion**
  Increase marketing efforts to acquire customers in **Low, Medium,** and **High** income segments to diversify the customer base.

Tools and Skills Demonstrated

Tools Used:

- Excel – Data Cleaning and Data Preparation
- Power BI – Data Modelling and Dashboard Visualizations
- DAX – Calculation of key performance metrics

Skills Demonstrated:

- Customer Segmentation Analysis
- RFM (Recency, Frequency, Monetary) Analysis
- Data Visualization and Dashboard design
- Business Insight generation
- Revenue Risk identification

For detailed and interactive analysis, please refer to the Power BI (.pbix) file provided in the project files.